Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100114 | *Subject: Basic Electronic Engineering* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) FET is a
(i) Voltage-controlled device (iii) Power device
(ii) Current-controlled device (iv) None

(b) An op-amp voltage follower has a gain
(i) 0 (iii) 1
(ii) 10 (iv) ∞

(c) Inverting amplifier with $R_1 = 10\ k\Omega, R_f = 100\ k\Omega$, gain =
(i) -1 (iii) -10
(ii) -5 (iv) -100

(d) The built-in potential of a diode increases when
(i) Doping increases (iii) Diode area decreases
(ii) Temperature increases (iv) Reverse bias increases

(e) BJT acts as a switch when in
(i) Cut-off (iii) Both
(ii) Saturation (iv) None

(f) A transistor has $I_c = 4\ mA\ and\ \beta = 100.\quad I_B =$.....
(i) $0.04\ \mu A$ (iii) $20\ \mu A$
(ii) $40\ \mu A$ (iv) $4\ \mu A$

(g) Reverse bias applied $voltage = 10\ V$, depletion $width = 2\ \mu m$. If reverse voltage becomes 40 V, the depletion width is approximately
(i) $2\ \mu m$ (iii) $8\ \mu m$
(ii) $4\ \mu m$ (iv) $16\ \mu m$

(h) The Zener diode in the regulator works under
(i) Forward bias (iii) Reverse non-breakdown
(ii) Reverse breakdown (iv) No bias

(i) 1's complement of 10110 =.....
(i) 01001 (iii) 10101
(ii) 01010 (iv) 11111

(j) The trans conductance 'g_m' of a JFET is equal to .....
(i) $\frac{-2I_{DSS}}{V_p}$ (iii) $\frac{-2I_{DSS}}{V_p}\left\{1-\frac{V_{GS}}{V_p}\right\}$
(ii) $\frac{2}{|V_p|}\sqrt{I_{DSS} * I_D}$ (iv) $\frac{I_{DSS}}{V_p}\left\{1-\frac{V_{GS}}{V_p}\right\}$

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Q.2	(a)	Describe the small-signal hybrid-π model. Derive expressions for $g_m, r_\pi, and\ r_o$.	[7]
	(b)	Explain the working of a voltage doubler circuit.	[7]
Q.3	(a)	To convert the hexadecimal number $(78AB)_{16}$ into base-5$(\ldots\ldots\ldots)_5$.	[7]
	(b)	A germanium diode carries a current of 1mA at room temperature when a forward bias of 0.15V is applied. Find the reverse saturation current at room temperature. Assume $\eta = 1$ for the Ge diode.	[7]
Q.4	(a)	Explain the effect of temperature on the V-I characteristics of a diode.	[7]
	(b)	Derive the gain expression for a non-inverting op-amp amplifier and discuss the case for a voltage follower.	[7]
Q.5	(a)	Explain the construction and working of the PNP BJT.	[7]
	(b)	Explain the operation of an enhancement-type MOSFET.	[7]
Q.6	(a)	Explain the important applications of a diode.	[7]
	(b)	What is the biasing of a BJT? Why is it necessary? Explain various methods of biasing.	[7]
Q.7	(a)	Design an EXOR gate using NAND gates only.	[7]
	(b)	Explain the zero-crossing detector and comparator using op-amps.	[7]
Q.8	(a)	Explain breakdown mechanisms (Zener & Avalanche).	[7]
	(b)	Explain the working of a full-wave bridge rectifier.	[7]
Q.9		Write short note on ***any two*** of the following:	[7 x 2 = 14]
	(a)	Pinch-off voltage and threshold voltage	
	(b)	VI characteristics of LED, Photodiode, and Solar cell	
	(c)	Direct and indirect band-gap semiconductors	
	(d)	Design a half adder and a full adder using logic gates	

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